
Supplement to “Extrapolation Is an Assumption: Sharp Limits for Pass@k Forecasting”

1 Identification Proofs

1.1 Observable information

Let

$$B_{s,b}(p) = \binom{b}{s} p^s (1-p)^{b-s}, \quad s = 0, \dots, b,$$

be the degree- b Bernstein basis. If $S \mid P = p$ is binomial, then $\Pr(S = s) = \int B_{s,b}(p) dF(p)$.

Proof of Proposition 1. The $b+1$ Bernstein polynomials form a basis for $\mathcal{P}_b$. Hence the count probabilities determine the integral of every polynomial in $\mathcal{P}_b$. Conversely, every linear functional of the count probabilities is the expectation of a polynomial in their span, so no other linear functional of F is directly determined.

For $k \leq b$, conditional on $S = s$, the probability that a uniformly chosen subset of k observed attempts contains at least one success is

$$\hat{h}_k(s) = 1 - \frac{\binom{b-s}{k}}{\binom{b}{k}},$$

with the ratio defined as zero when $b-s < k$. Conditional exchangeability gives $\mathbb{E}_p[\hat{h}_k(S)] = 1 - (1-p)^k$. Averaging over tasks and F proves unbiasedness.

For $k > b$, $h_k(p) = 1 - (1-p)^k$ has degree k and is not in $\mathcal{P}_b$. Theorem 1 constructs two laws that agree on every element of $\mathcal{P}_b$ but disagree on h_k , proving nonidentification in general. $\square$

1.2 The identified interval for fixed moments

For a feasible moment vector $\mu = (\mu_0, \dots, \mu_b)$, define

$$\mathcal{F}(\mu) = \left\{ F : \int p^j dF(p) = \mu_j, \quad j = 0, \dots, b \right\}.$$

The set is weakly compact because probability measures on $[0, 1]$ are weakly compact and the moment constraints are continuous. The functional $F \mapsto \int h_k dF$ is continuous, so its minimum and maximum over $\mathcal{F}(\mu)$

exist. Convexity implies that every value between the two endpoints is attained.

The lower primal problem and its generalized moment dual are

$$\inf_{F \in \mathcal{F}(\mu)} \int h_k dF \quad \text{and} \quad \sup_{q \in \mathcal{P}_b} \left\{ \sum_{j=0}^b q_j \mu_j : q \leq h_k \right\}.$$

The upper pair reverses the optimization and the envelope. Feasibility and compact support give equality of primal and dual values (de Klerk and Laurent, 2019; Halaseh et al., 2026). A dual envelope need not attain its optimum when μ lies on the boundary of the moment cone. This is why the main paper uses a supremum and an infimum.

Because $h_k - q$ and $q - h_k$ are univariate polynomials, their nonnegativity on $[0, 1]$ can also be represented exactly by the Markov–Lukacs certificate. If a polynomial r has even degree $2d$,

$$r(p) = \sigma_0(p) + p(1-p)\sigma_1(p),$$

where σ_0, σ_1 are sums of squares of degrees at most $2d$ and $2d-2$. If its degree is odd $2d+1$,

$$r(p) = p\sigma_0(p) + (1-p)\sigma_1(p),$$

where both sums of squares have degree at most $2d$. Gram-matrix representations turn the endpoint problems into finite semidefinite programs. Their size grows with k , so our largest-horizon computations instead use linear programs on Chebyshev grids and monotone outer projection.

1.3 Sharp worst-case diameter

Proof of Theorem 1. Set $X = 1 - P$. Matching the first b moments of P is equivalent to matching those of X . Let F, G be any two such laws and let $q \in \mathcal{P}_b$.

Since $\int q d(F - G) = 0$,

$$\begin{aligned} |\theta_k(F) - \theta_k(G)| &= \left| \int x^k d(F - G)(x) \right| \\ &= \left| \int (x^k - q(x)) d(F - G)(x) \right| \\ &\leq 2\|x^k - q\|_\infty. \end{aligned}$$

Taking the infimum over q proves $D_{b,k} \leq 2E_{b,k}$.

For the reverse inequality, let $C = C[0, 1]$ with the uniform norm, and consider the quotient Banach space $C/\mathcal{P}_b$. The norm of the coset $[x^k]$ is its distance to $\mathcal{P}_b$, namely $E_{b,k}$. By Hahn–Banach, there is a continuous linear functional Λ on C with operator norm one such that

$$\Lambda(q) = 0 \quad (q \in \mathcal{P}_b), \quad \Lambda(x^k) = E_{b,k}.$$

The Riesz representation theorem gives a finite signed measure ν with total variation one and $\Lambda(f) = \int f d\nu$. Since the constant function lies in $\mathcal{P}_b$, $\nu([0, 1]) = 0$. In the Jordan decomposition $\nu = \nu^+ - \nu^-$, both parts therefore have mass $1/2$.

Define probability measures $F^* = 2\nu^+$ and $G^* = 2\nu^-$. They match every polynomial of degree at most b , and

$$\int x^k d(F^* - G^*) = 2E_{b,k}.$$

Their count laws match by the Bernstein representation. Their pass@ k values have the same absolute gap because the constant part of $1 - x^k$ cancels. Thus $D_{b,k} \geq 2E_{b,k}$. $\square$

1.4 Newman–Rivlin mapping

Proof of Corollary 1. Let $E_n^{[-1, 1]}(f)$ denote the best degree- n uniform approximation on $[-1, 1]$. If $q \in \mathcal{P}_b$, then $q(z^2)$ is an even degree- $2b$ polynomial and

$$\sup_{z \in [-1, 1]} |z^{2k} - q(z^2)| = \sup_{x \in [0, 1]} |x^k - q(x)|.$$

Conversely, symmetrizing any degree- $2b$ approximant to the even function z^{2k} cannot increase its error. The resulting even polynomial is $q(z^2)$ for a $q \in \mathcal{P}_b$. Hence

$$E_{b,k} = E_{2b}^{[-1, 1]}(z^{2k}).$$

Theorem 3 of Newman and Rivlin (1976) states that

$$\frac{1}{4e} P_{n,m} \leq E_m^{[-1, 1]}(z^n) \leq P_{n,m},$$

where $P_{n,m}$ is the probability that the absolute difference between heads and tails in n fair tosses exceeds m . With $n = 2k, m = 2b$, that event is $|H - k| > b$

for $H \sim \text{Binomial}(2k, 1/2)$. Multiplication by two and Theorem 1 give the displayed bounds.

For large k , the standardized threshold is $b/\sqrt{k/2} = \sqrt{2}b/\sqrt{k}$. The central limit theorem therefore places the transition at $b/\sqrt{k}$ of constant order. If $b/\sqrt{k} \rightarrow \infty$, the tail and diameter vanish. If that ratio remains bounded, the lower bound stays away from zero along nondegenerate limits. $\square$

1.5 Declared support restrictions

Proof of Corollary 2. Under $P \in [p_0, 1]$, set $X = 1 - P \in [0, a]$ with $a = 1 - p_0$. If $a = 0$, the target is constant and the claim is immediate. Otherwise write $X = aY$. The map from laws of X to laws of $Y \in [0, 1]$ is bijective. Matching degree- b moments is preserved under this scaling, and

$$|\mathbb{E}_F X^k - \mathbb{E}_G X^k| = a^k |\mathbb{E}_F Y^k - \mathbb{E}_G Y^k|.$$

Taking the supremum over matching laws and applying Theorem 1 gives $D_{b,k}(p_0) = a^k D_{b,k}$. $\square$

2 Confidence-Set Lower Bounds

2.1 Exactly matching count laws

First part of Theorem 2. Let $A_F = \{\theta_k(F) \in C\}$ and $A_G = \{\theta_k(G) \in C\}$. Under a shared data law,

$$\Pr(A_F \cap A_G) \geq 1 - \Pr(A_F^c) - \Pr(A_G^c) \geq 1 - 2\alpha.$$

On the intersection, C contains two points separated by Δ , so its diameter is at least Δ . The expected-length claim follows from $\mathbb{E}[\text{diam}(C)] \geq \Delta \Pr\{\text{diam}(C) \geq \Delta\}$. $\square$

2.2 Nearby finite-task experiments

Lemma 1 (Affinity bound). *For probability mass functions q, r , define $\rho(q, r) = \sum_s \sqrt{q_s r_s}$. Then*

$$\text{TV}(q, r) \leq \sqrt{1 - \rho(q, r)^2}.$$

For M iid observations, $\rho(q^M, r^M) = \rho(q, r)^M$.

Proof. Cauchy–Schwarz gives

$$\begin{aligned} \text{TV}(q, r) &= \frac{1}{2} \sum_s |\sqrt{q_s} - \sqrt{r_s}| (\sqrt{q_s} + \sqrt{r_s}) \\ &\leq \frac{1}{2} \sqrt{\sum_s (\sqrt{q_s} - \sqrt{r_s})^2} \sqrt{\sum_s (\sqrt{q_s} + \sqrt{r_s})^2} \\ &= \sqrt{(1 - \rho(q, r))(1 + \rho(q, r))}. \end{aligned}$$

The product identity follows by factorizing the sum over product outcomes. $\square$

Finite-task part of Theorem 2. Let P_F, P_G denote the two full data laws. Coverage and total variation give

$$\begin{aligned} P_F(A_G) &\geq P_G(A_G) - \text{TV}(P_F, P_G) \\ &\geq 1 - \alpha - \text{TV}(P_F, P_G). \end{aligned}$$

Combining this with $P_F(A_F) \geq 1 - \alpha$ yields

$$P_F(A_F \cap A_G) \geq 1 - 2\alpha - \text{TV}(P_F, P_G).$$

The preceding lemma bounds the final term by $\sqrt{1 - \rho(q_F, q_G)^{2M}}$. On the intersection the interval spans the target gap. $\square$

2.3 Convex modulus program and repair

On support points $p_1, \dots, p_J$, let w, z be two vectors in the probability simplex. Define the $(b+1) \times J$ matrix

$$A_{sj} = \binom{b}{s} p_j^s (1 - p_j)^{b-s}$$

and $t_j = h_k(p_j)$. For a desired product-TV budget v , the grid-restricted program is

$$\begin{aligned} &\text{maximize} \quad t^\top (w - z) \\ &\text{subject to} \quad w, z \geq 0, \quad \mathbf{1}^\top w = \mathbf{1}^\top z = 1, \\ &\quad \sum_{s=0}^b \sqrt{(Aw)_s (Az)_s} \geq (1 - v^2)^{1/(2M)}. \end{aligned}$$

The affinity is concave in (w, z) , so its superlevel set is convex. The objective is linear. Any feasible solution is an actual pair of mixing laws, and therefore a valid lower-bound witness for the unrestricted modulus.

Numerical solvers can return weights with tiny negative entries or affinity slightly below the requested value. We first clip negative weights and renormalize. Let $m = (w+z)/2$ and $d = (w-z)/2$. We then replace the pair by

$$w_\lambda = m + \lambda d, \quad z_\lambda = m - \lambda d, \quad 0 \leq \lambda \leq 1.$$

At $\lambda = 0$, the two laws coincide and have affinity one. We use bisection to select the largest feasible λ , followed by a relative 10^{-10} shrinkage. Every reported target gap is recomputed from this repaired pair. Thus solver optimality labels do not affect validity of the lower bounds.

2.4 Adaptive policies

Proof of Proposition 2. Condition on the policy's external random seed, so its decisions are deterministic functions of the observed transcript. Consider any finite transcript compatible with the policy. For task

i , suppose it contains $t_i \leq b$ observations with s_i successes. Integrating its latent probability gives the history probability

$$\int p^{s_i} (1-p)^{t_i-s_i} dF(p).$$

The integrand has degree $t_i \leq b$, so this value is unchanged when F is replaced by a law G with matching first b moments. Latent task probabilities are independent, hence the probability of the collection of task histories is the product of these integrals. Policy decisions add only indicators that the transcript follows the deterministic rule. Therefore every complete transcript has the same probability under F and G . Averaging over the external random seed proves the claim for a randomized policy. $\square$

3 Honest Interval Constructions

3.1 Global Bernstein interval

Theorem 1. Fix arbitrary $p_1, \dots, p_M \in [0, 1]$, and independently draw $S_i \sim \text{Binomial}(b, p_i)$. If $\|q - h_k\|_\infty \leq \epsilon$, where

$$q(p) = \sum_{s=0}^b a_s B_{s,b}(p),$$

and $R = \max_s a_s - \min_s a_s$, then the global polynomial interval in the main paper covers

$$\theta_{M,k} = \frac{1}{M} \sum_{i=1}^M h_k(p_i)$$

with probability at least $1 - \alpha$. If instead P_i are iid from F , the same interval also covers population $\theta_k(F)$ with that probability.

Proof. Let $\bar{a} = M^{-1} \sum_i a_{S_i}$. The Bernstein identity gives

$$\mathbb{E}[\bar{a}] = \frac{1}{M} \sum_i q(p_i).$$

The deterministic approximation error therefore satisfies

$$|\mathbb{E}[\bar{a}] - \theta_{M,k}| \leq \epsilon.$$

Each a_{S_i} lies in an interval of length R . Hoeffding's inequality for independent, not necessarily identically distributed variables yields

$$\Pr\{|\bar{a} - \mathbb{E}\bar{a}| > Rc_M\} \leq 2 \exp(-2Mc_M^2) = \alpha.$$

The triangle inequality gives coverage before clipping, and intersecting with $[0, 1]$ cannot remove the target.

For iid tasks, the mixture counts S_i are iid and $\mathbb{E}[a_{S_i}] = \int q(p) dF(p)$. Repeating the same approximation and concentration argument gives population coverage. $\square$

Table 1: Finite-task modulus witnesses for $M = 128, b = 78, \alpha = 0.05$. All displayed product-TV bounds are at most the requested v .

k	v	target gap	span probability	expected-length lower	solver label
100	.05	.00531	.850	.00451	optimal-inaccurate
100	.10	.01063	.800	.00850	optimal
100	.20	.02139	.700	.01497	optimal
100	.30	.03246	.600	.01948	optimal
100	.40	.04407	.500	.02203	optimal
1,000	.05	.28233	.850	.23998	optimal-inaccurate
1,000	.10	.33187	.800	.26549	optimal-inaccurate
1,000	.20	.39457	.700	.27620	optimal-inaccurate
1,000	.30	.44060	.600	.26436	optimal-inaccurate
1,000	.40	.47946	.500	.23973	optimal-inaccurate
10,000	.05	.80759	.850	.68645	optimal-inaccurate
10,000	.10	.83681	.800	.66945	optimal-inaccurate
10,000	.20	.86959	.700	.60871	optimal-inaccurate
10,000	.30	.89134	.600	.53480	optimal-inaccurate
10,000	.40	.90786	.500	.45393	optimal-inaccurate

Design program. On evaluation points u_ℓ , the implemented linear program minimizes

$$\epsilon + c_M(a_{\max} - a_{\min})$$

over $a_0, \dots, a_b, \epsilon, a_{\min}, a_{\max}$, subject to

$$|q(u_\ell) - h_k(u_\ell)| \leq \epsilon, \quad a_{\min} \leq a_s \leq a_{\max}.$$

After solving, we certify the continuous error. The Bernstein derivative identity gives

$$q'(p) = b \sum_{s=0}^{b-1} (a_{s+1} - a_s) B_{s,b-1}(p),$$

so $|q'(p)| \leq b\Delta_a$, where $\Delta_a = \max_s |a_{s+1} - a_s|$. Also $h'_k(p) = k(1-p)^{k-1}$, which decreases in p . On adjacent grid points $u_\ell, u_{\ell+1}$, every p is within half the interval width of an endpoint. Hence the continuous error on that interval is at most

$$\max\{|q(u_\ell) - h_k(u_\ell)|, |q(u_{\ell+1}) - h_k(u_{\ell+1})|\} + \frac{u_{\ell+1} - u_\ell}{2} [k(1 - u_\ell)^{k-1} + b\Delta_a].$$

We use the maximum of these bounds plus 10^{-10} as ϵ .

Asymptotic sharpness. In the ideal continuous design problem, every feasible polynomial has $\epsilon \geq E_{b,k}$. Conversely, for any $\eta > 0$, choose a polynomial with error at most $E_{b,k} + \eta$. Its finite Bernstein coefficient range is multiplied by $c_M \rightarrow 0$. Therefore the optimized half-width converges to $E_{b,k}$, and the full worst-case width converges to $2E_{b,k} = D_{b,k}$.

3.2 Count-CDF projection interval

For thresholds $s = 0, \dots, b-1$, write

$$g_s(p) = \Pr\{\text{Binomial}(b, p) \leq s\}.$$

This function is decreasing in p . The pass target $h_k(p)$ is increasing.

Theorem 2. For iid tasks from F , use radius

$$\delta_{\text{iid}} = \sqrt{\frac{\log(2/\alpha)}{2M}}.$$

For fixed $p_1, \dots, p_M$ with independent counts, use

$$\delta_{\text{fix}} = \sqrt{\frac{\log(2b/\alpha)}{2M}}.$$

The endpoint-envelope linear program in the main paper gives a $(1 - \alpha)$ outer confidence interval for $\theta_k(F)$ in the iid case, and for $\theta_{M,k}$ in the fixed case.

Proof. In the iid case, the counts S_i are iid draws from the mixture count CDF $G_s = \int g_s(p) dF(p)$. The DKW inequality (Massart, 1990) gives

$$\Pr\left\{\max_s |\hat{G}_s - G_s| > \delta_{\text{iid}}\right\} \leq \alpha.$$

In the fixed case, $\mathbf{1}\{S_i \leq s\}$ are independent variables in $[0, 1]$, with average expectation

$$G_{M,s} = \frac{1}{M} \sum_i g_s(p_i).$$

For each threshold, Hoeffding’s inequality gives a two-sided error probability at most $2 \exp(-2M\delta^2)$. A union bound over the b nontrivial thresholds gives the stated radius.

It remains to prove projection validity on either simultaneous-band event. Let $F_\star$ be F in the iid setting and the empirical law $M^{-1} \sum_i \delta_{p_i}$ in the fixed setting. On a partition $[e_j, e_{j+1}]$, set $w_j = F_\star([e_j, e_{j+1}])$,

Table 2: Worst-case population identification. Grid-based approximation and moment-pair programs agree at the displayed precision.

b	k	approx. $D_{b,k}$	max moment resid.
4	128	.6677	2.1×10^{-15}
8	1,000	.7833	5.7×10^{-12}
16	1,000	.5086	5.1×10^{-15}
32	10,000	.7066	4.3×10^{-12}
64	10,000	.3886	1.6×10^{-14}
78	1,000	.000489	8.2×10^{-11}
78	10,000	.28495	7.3×10^{-13}

with a fixed convention at shared endpoints. Monotonicity gives

$$\sum_j w_j g_s(e_{j+1}) \leq \int g_s dF_\star \leq \sum_j w_j g_s(e_j).$$

Combining these inequalities with the confidence band shows that the true bin masses are feasible for the linear constraints. A second use of monotonicity gives

$$\sum_j w_j h_k(e_j) \leq \int h_k dF_\star \leq \sum_j w_j h_k(e_{j+1}).$$

The minimized lower envelope can only be smaller than the left side, and the maximized upper envelope can only be larger than the right side. The projected interval therefore contains the target on the simultaneous-band event. $\square$

Numerical conditioning. To compute the upper endpoint, maximizing h_k is implemented as minimizing the failure probability $(1-p)^k$, then subtracting the optimum from one. This avoids an objective numerically indistinguishable from the constant one at large k . Each endpoint first uses HiGHS, then retries its dual simplex and interior-point methods at a relaxed 10^{-8} feasibility tolerance. Every reported run completed successfully after this reparameterization.

4 Numerical Identification Results

The $b = 78, k = 100$ error is below double-precision resolution. We do not use the numerical zero as a theorem. Corollary 1 gives the rigorous bound

$$3.58 \times 10^{-33} \leq D_{78,100} \leq 3.89 \times 10^{-32}.$$

At full precision, the two induced count probability vectors differ by at most 1.7×10^{-12} . Their target values are 0.9501871171 and 0.1668880229. Their stored degree- b Chebyshev moment residual is 5.7×10^{-12} . For this ambiguity example, the numerical primal and dual diameters differ by less than 10^{-11} .

5 Experiment Protocols and Additional Results

5.1 Synthetic interval audit

Each repetition first samples 128 latent task probabilities from the specified discrete or Beta law. It then samples one binomial count per task. Coverage is evaluated against the realized finite-task target

$$\theta_{M,k} = M^{-1} \sum_{i=1}^M [1 - (1 - p_i)^k].$$

This makes the comparison conditional on the sampled benchmark and matches the fixed-task guarantee of both honest intervals.

The exact pair is the law in Table 3. Finite-pair laws are the repaired $v = 0.2$ modulus witnesses for $b = 78, k \in \{1000, 10000\}$. The benign laws are Beta(1, 1) and Beta(0.35, 5), discretized by 20,001 equal-mass midpoint quantiles for simulation.

For each taskwise Bayesian interval, the posterior is

$$p_i \mid S_i \sim \text{Beta}(S_i + 1, b - S_i + 1).$$

We use 2,000 independent joint posterior draws, compute the task-average pass@ k for each draw, and take its 2.5% and 97.5% quantiles. The posterior mean is computed analytically. Population Beta point estimates use a ten-start Beta-binomial maximum-likelihood fit in log parameters. We also report the direct plug-in and the clipped binomial smoothed Good-Toulmin point estimates.

The honest intervals’ empirical coverage of one is consistent with their guarantees but should not be interpreted as exact calibration. Their concentration radii are conservative at $M = 128$. The comparison of widths is the relevant positive result: the projection interval adapts substantially under Uniform and rare Beta laws, while retaining the required width on hard pairs.

5.2 Public response-bank audit

Data provenance. We use the correctness indicators from all 22 configurations of `ScalingIntelligence/monkey_business`, the response-bank release of Brown et al. (2024). The Hugging Face dataset revision observed during the audit was `a9f8f73bcd6948a57ed922cba4e48062ef95f553`. Its dataset card declares the MIT license. We fetch only the `is_corrects` and original task index columns from the hosted Parquet shards, sort by original task index, and store Boolean arrays plus source URLs. The 22 panels contain 127–140 tasks and 10,000 attempts per task.

Table 3: Atomic laws for the $b = 8, k = 1000$ ambiguity example. Weights shown here are rounded. The artifact stores full precision.

High pass@1000 law		Low pass@1000 law	
p	weight	p	weight
.002934	.940617	.000000	.833112
.149794	.038257	.041906	.134095
.501963	.011496	.311564	.018474
.854108	.006747	.692430	.008329
1.000000	.002882	.962090	.005991

Table 4: Synthetic 95% interval audit over 400 repetitions per scenario. “Task B.” is taskwise Beta, “Projection” is fixed-task count-CDF projection, and “Global” is the Bernstein interval.

Scenario	Method	coverage	mean width	MAE	repetitions
Uniform, $b = 78, k = 1000$	Task B.	.9625	.0059	.0011	400
	Projection	1.0000	.1901	.0941	400
	Global	1.0000	.7523	.2606	400
Rare Beta, $b = 78, k = 1000$	Task B.	.0000	.0379	.1233	400
	Projection	1.0000	.5470	.1224	400
	Global	1.0000	.8056	.1637	400
Exact low, $b = 8, k = 1000$	Task B.	.0000	.0210	.8235	400
	Projection	1.0000	1.0000	.3318	400
	Global	1.0000	1.0000	.3318	400
Exact high, $b = 8, k = 1000$	Task B.	.0000	.0210	.0415	400
	Projection	1.0000	.9999	.4502	400
	Global	1.0000	1.0000	.4503	400
Finite first, $b = 78, k = 1000$	Task B.	.0000	.0549	.1384	400
	Projection	1.0000	.9472	.2792	400
	Global	1.0000	.9531	.2670	400
Finite second, $b = 78, k = 1000$	Task B.	.0000	.0551	.5350	400
	Projection	1.0000	.9490	.1165	400
	Global	1.0000	.9540	.1285	400
Finite first, $b = 78, k = 10000$	Task B.	.0000	.0198	.0204	400
	Projection	1.0000	1.0000	.4722	400
	Global	1.0000	1.0000	.4722	400
Finite second, $b = 78, k = 10000$	Task B.	.0000	.0198	.8900	400
	Projection	1.0000	1.0000	.3974	400
	Global	1.0000	1.0000	.3974	400

References and resampling. For a response bank with $N = 10000$ attempts and c_i successful attempts on task i , the finite-bank pass reference is

$$\hat{\theta}_k = \frac{1}{M} \sum_{i=1}^M \left[1 - \frac{\binom{N-c_i}{k}}{\binom{N}{k}} \right].$$

This is the expected without-replacement pass@ k of a uniformly sampled subset from the bank. For each of 50 seeds, we independently shuffle each task’s responses. Equal allocation uses the first $\lfloor B/M \rfloor$ responses per task. Dynamic allocation implements Algorithm 1 of Kazdan et al. (2025): it initializes each task, then allocates attempts according to the prescribed function of failures and successes. The nominal budgets are 1,000, 3,000, and 10,000. Targets are 100, 1,000, and 10,000.

Population Beta fit. Given possibly unequal b_i and counts s_i , the likelihood is

$$\prod_i \binom{b_i}{s_i} \frac{B(s_i + \alpha, b_i - s_i + \beta)}{B(\alpha, \beta)}.$$

The combinatorial factors do not affect optimization. We optimize $\log \alpha, \log \beta$ with analytic gradients from ten starting points, using bounds $[-14, 18]$ on both log parameters. Boundary fits are retained and flagged. Forecasts use

$$1 - \frac{B(\alpha, \beta + k)}{B(\alpha, \beta)}.$$

All 59,400 forecast rows used in the main point-estimate audit have finite objectives. The artifact preserves convergence, gradient, and boundary diagnostics.

Table 5: Uniform-allocation public point forecasts at total budget 10,000. Each row contains 1,100 panel-resamples.

k	method	MAE	RMSE	fraction $> .1$
100	Population Beta	.0175	.0225	.000
100	SGT	.0161	.0223	.000
100	Plug-in	.0396	.0516	.055
100	Taskwise Beta	.2901	.3451	.729
1,000	Population Beta	.0414	.0558	.092
1,000	SGT	.2000	.2764	.605
1,000	Plug-in	.1481	.1845	.511
1,000	Taskwise Beta	.3843	.4846	.682
10,000	Population Beta	.0697	.1300	.170
10,000	SGT	.3529	.4688	.713
10,000	Plug-in	.2365	.2869	.737
10,000	Taskwise Beta	.3326	.4489	.682

Table 6: Fixed-task projection audit over 22 panels and 50 resamples.

k	ref. inclusion	mean width	median width	width $> .9$
100	1.000	.371	.395	.000
1,000	1.000	.714	.787	.405
10,000	1.000	.721	.787	.405

Reference inclusion is not a frequentist coverage estimate. The full bank is finite, contains the sampled prefix, and estimates rather than equals a latent target. We report it as a reproducible descriptive check. The synthetic study provides the controlled coverage experiment.

5.3 Compute and software

All optimization and data experiments ran in Python 3.10 on Argonne Swing. CPU computations were submitted to a GPU-only PBS queue but do not use the allocated accelerator. Core packages are NumPy, SciPy 1.15.3, pandas, Matplotlib, CVXPY, Clarabel, PyArrow, and fsspec. Four spawned workers run the coverage and projection audits with all BLAS thread counts set to one.

The full 400-repetition synthetic audit took 2 minutes 41 seconds on one node. The 3,300-row public projection audit took 2 minutes 16 seconds. The artifact uses fixed seeds, writes outputs only after all worker results return, and contains ten deterministic tests. Tests cover the combinatorial estimator, dynamic allocation, Beta-binomial recovery, taskwise posterior computation, smoothed Good–Toulmin interpolation, identification primal-dual agreement, continuous bias certification, finite-modulus feasibility, and projection containment.

6 Additional Literature Boundary

The closest statistical model predates $\text{pass}@k$. Colwell et al. (2004) use the same incidence binomial mixture and accumulation functional. They recommend moment interpolation, fit finite mixtures for extrapolation, and caution that the asymptote is unreliable. Orlitsky et al. (2016) derive minimax smoothed Good–Toulmin estimators in both abundance and Bernoulli-product incidence models. Their estimable extrapolation factor grows logarithmically with sample size. Our work addresses a different quantity: the exact finite- b identified diameter, finite-task interval lower bounds, and honest $\text{pass}@k$ projection.

Mao and Lindsay (2007) show that unseen-class odds in a zero-truncated Poisson mixture are discontinuous and that uniformly informative two-sided inference for total richness is impossible. That target is unbounded and asymptotic. Our finite-horizon target remains in $[0, 1]$, producing a bounded but nontrivial ambiguity that changes continuously with b and k . Daley and Smith (2013) use rational continuation for sequencing-library complexity, while Shestopaloff (2025) fits parametric accumulation-rate tails and uses a parametric bootstrap. Both demonstrate that structural continuation can be practically effective.

Basu et al. (2026) give exact pointwise CDF and quantile intervals for a binomial mixing distribution by test inversion. They explicitly discuss the finite-moment identification boundary. They do not study the $\text{pass}@k$ functional, its sharp diameter, or the b^2 horizon. Kaido et al. (2019) provide generic asymptotically calibrated projection intervals for partially identified moment models. Our count-CDF interval is a specialized nonasymptotic outer projection whose validity follows from DKW or finite Hoeffding bands and monotone bin endpoints.

Recent $\text{pass}@k$ papers impose useful structure rather than study nonparametric identification. Schaeffer et al. (2025) explain aggregate power laws through heavy lower tails. Levi (2025) derive the same curve under a Beta failure model, while Levi (2026) connect a regularly varying difficulty tail to training-dependent inference scaling. Kazdan et al. (2025) propose population Beta-binomial fitting and dynamic allocation. Anonymous Authors (2026) propagate taskwise Beta or Dirichlet posteriors through richer repeated-sampling profiles. Hariri et al. (2026) use taskwise Bayesian inference to stabilize model evaluation. Our audits target the extrapolative assumptions in these population and taskwise models, not their within-model posterior algebra. A separate diagnostic by Zhou et al. (2026) shows that a deterministic intervention can reach some tasks missed by ordinary

sampling. That result concerns the choice of decoding regime, not extrapolation from a fixed regime’s finite count law.

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